

ASEN-6062
Comp HWK #2
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Create restricted 4 body system
add 'massless' body to 3BP.

$\vec{R}_3$ measured from com

repeat 1a,b,c focused on 4th body
 $\vec{R}_0 = \vec{0}$

for b & c plot perturbed & unperturbed trajectory
provide EoM for body 4 & plot trajectories

CR4BP

3a) Verify conservation residuals magnitude change w/ tolerance
 $m_1 + m_2 + m_3 = 1$

$$\vec{R}_1 = \vec{R}_{12} = \vec{R}_2 - \vec{r}_1$$

$$\vec{R}_2 = \frac{1}{m_1 + m_2} [m_1(\vec{r}_3 - \vec{r}_1) + m_2(\vec{r}_3 - \vec{r}_2)]$$

$$\vec{R}_N = \vec{R}_C$$

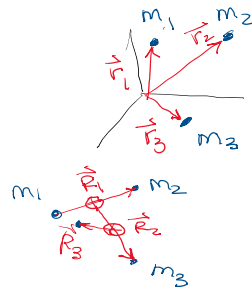
$$\vec{R}_{N-1} = \frac{1}{\sum_{i=1}^{N-1} m_i} \sum_{i=1}^{N-1} m_i \vec{r}_{iN}$$

use Euler w/ unequal mass

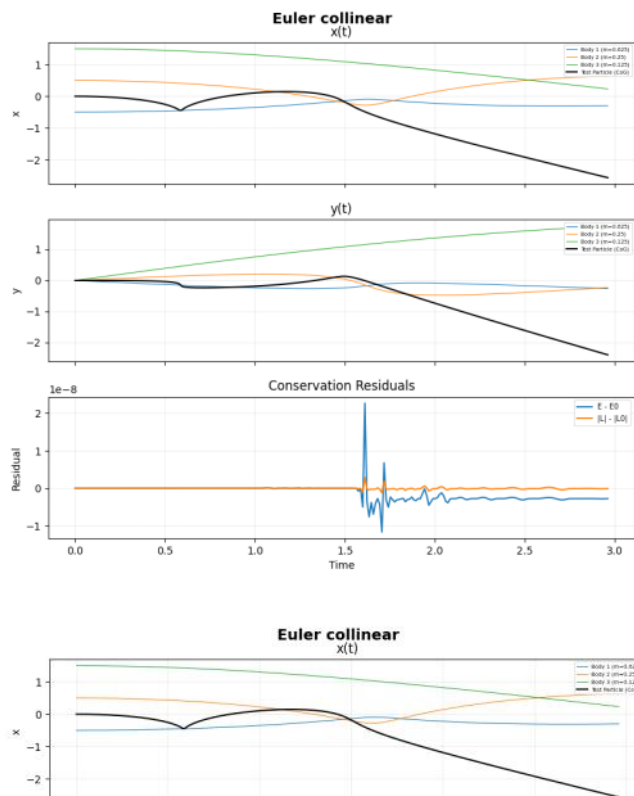
$$M = [\frac{5}{8} \quad \frac{1}{4} \quad \frac{1}{8}]$$

$$\vec{r} = [-a, 0, a]$$

com
 m_1 m_2 m_3

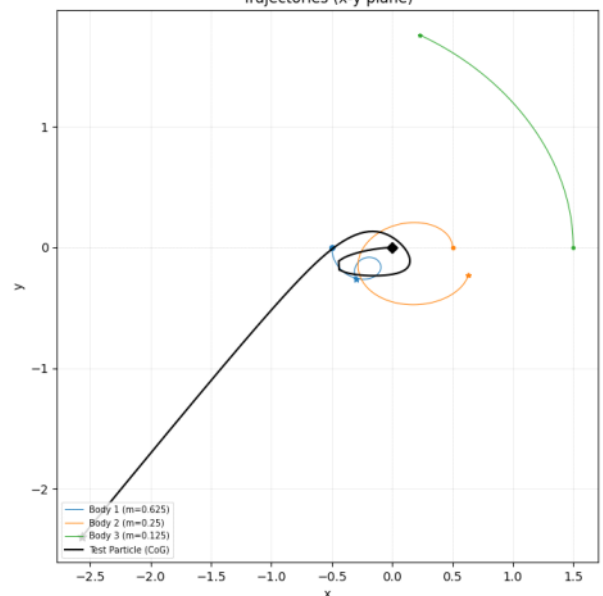


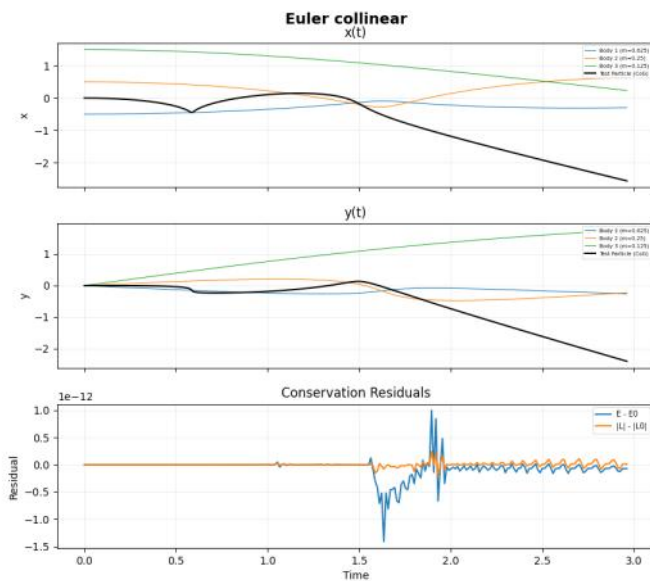
Magnitude of errors in conserved values
align w/ tolerance of integrator:



Euler collinear

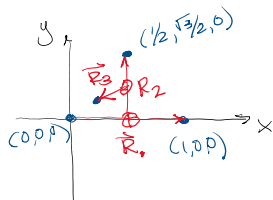
Trajectories (x-y plane)





-2.5 -2.0 -1.5 -1.0 -0.5 0.0 0.5 1.0 1.5
x

3b)



$$\mu_1 = \frac{m_1 m_2}{m_1 + m_2}$$

$$\mu_2 = \frac{(m_1 + m_2) m_3}{m_1 + m_2 + m_3}$$

$$\ddot{\vec{r}} = G \sum_{i=1}^3 m_i \frac{(\vec{r}_4 - \vec{r}_i)}{|\vec{r}_4 - \vec{r}_i|^3}$$

$$L = T - U$$

$$\mu_i \ddot{\vec{r}}_i = - \frac{\partial U}{\partial \vec{r}_i}$$

$$\vec{x} = \begin{bmatrix} \vec{r} \\ \vec{p} \end{bmatrix}$$

$$\ddot{\vec{r}} = \frac{1}{\mu_i} \left(- \frac{\partial U}{\partial \vec{r}_i} \right)$$

$$U = -G \sum_{i,j} \frac{m_i m_j}{|\vec{r}_{ij}|}$$

$$\frac{\partial U}{\partial \vec{r}_i} = -G \sum_j \frac{m_i m_j}{|\vec{r}_{ij}|^3} \vec{r}_{ij} \frac{\partial \vec{r}_{ij}}{\partial \vec{r}_i}$$

$$\frac{\partial \vec{r}_{ij}}{\partial \vec{r}_i} = \begin{cases} 0 & l > j-1 \\ u & l = j-1 \\ \frac{m_{i+l}}{m_{i+l-1}} & j-2 \leq l < i-1 \\ -\frac{m_{i-1}}{m_i} u & l = i-1 \\ 0 & l < i-1 \end{cases}$$

$\vec{R}_3$ is system COM

$$\vec{r}_4 = \vec{R}_3$$

in COM frame, $\vec{R}_3 = \vec{0}$

$$m = \begin{bmatrix} 1/2 & 1/3 & 1/6 \end{bmatrix} \quad M = 1$$

$$\vec{r}_1 = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$\vec{r}_2 = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

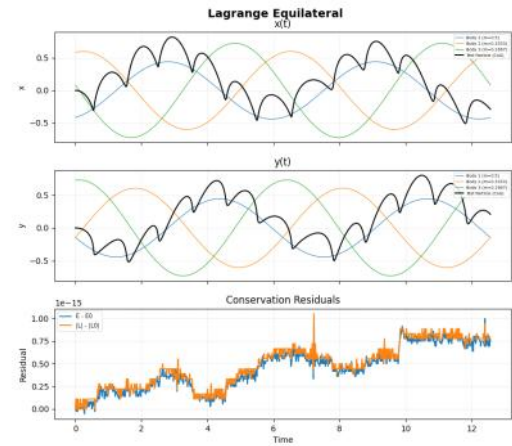
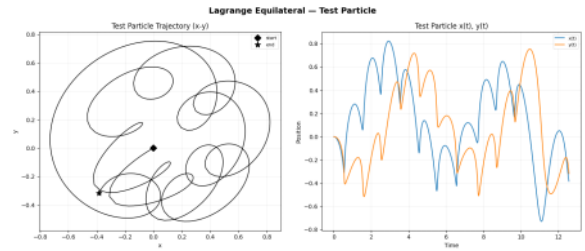
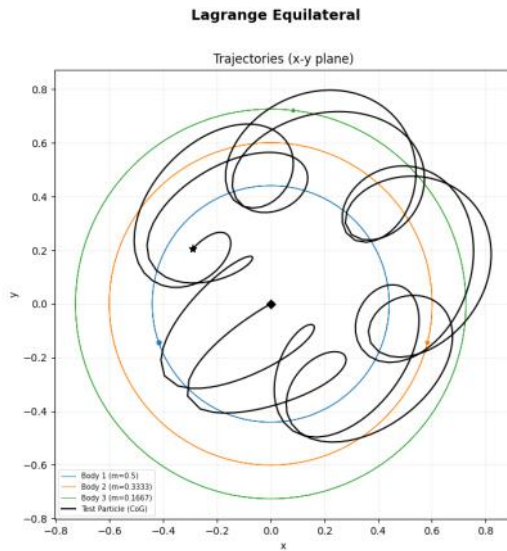
$$\vec{r}_3 = \begin{bmatrix} 1/2 & \sqrt{3}/2 & 0 \end{bmatrix}$$

$$\vec{r}_4 = \begin{bmatrix} 5/2 & \sqrt{3}/2 & 0 \end{bmatrix}$$

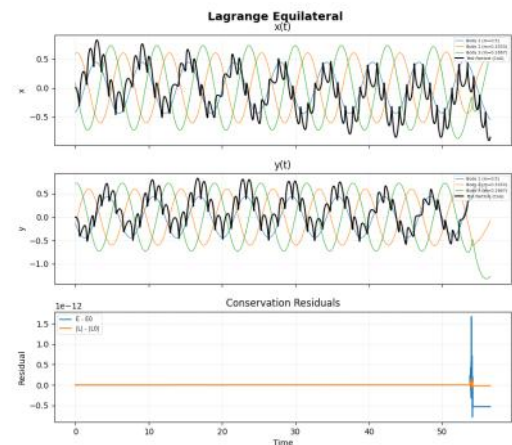
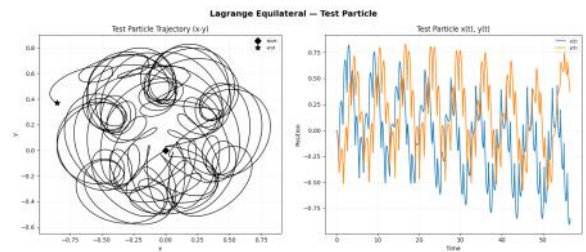
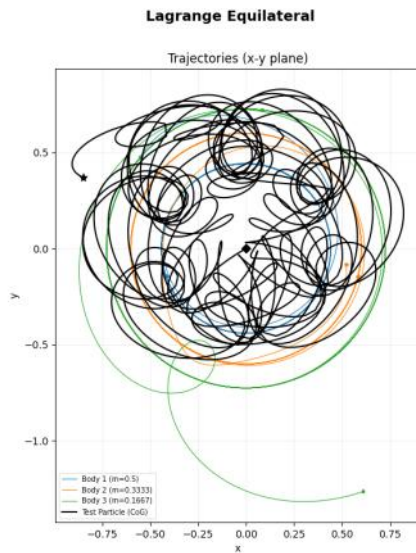
$$\dot{\vec{r}} = \omega \times \vec{r} \quad \omega = 1$$

Eqs of motion

This plot shows an unperturbed system. The fourth body stays in the system & does not escape.

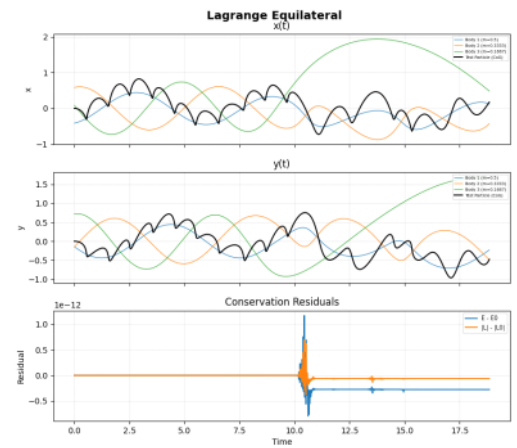
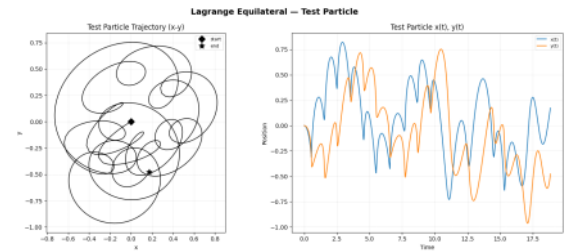
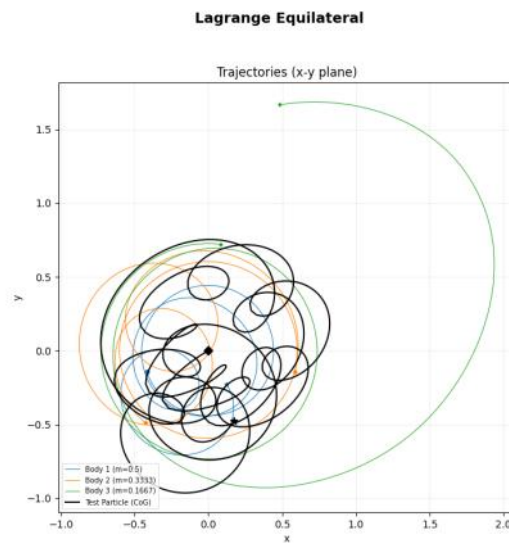


This plot shows 9 periods of the unperturbed system. The 3rd body escapes but the fourth body continues to be bound to the system.



Then a perturbation of 0.001 was introduced in the x & y position of the heaviest body. These plots show 3 periods of the orbits. The third body is now unbound.

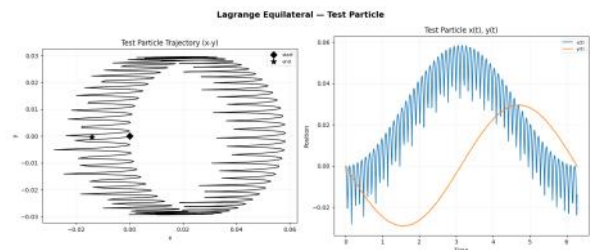
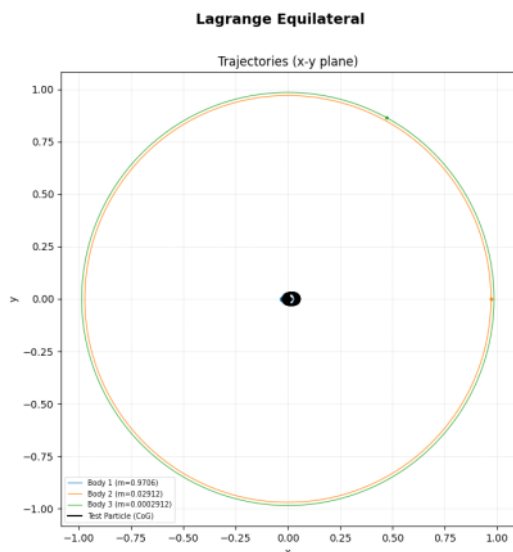
x & y position of the heaviest body. These plots show 3 periods of the orbits. The third body escapes significantly faster in the perturbed environment.



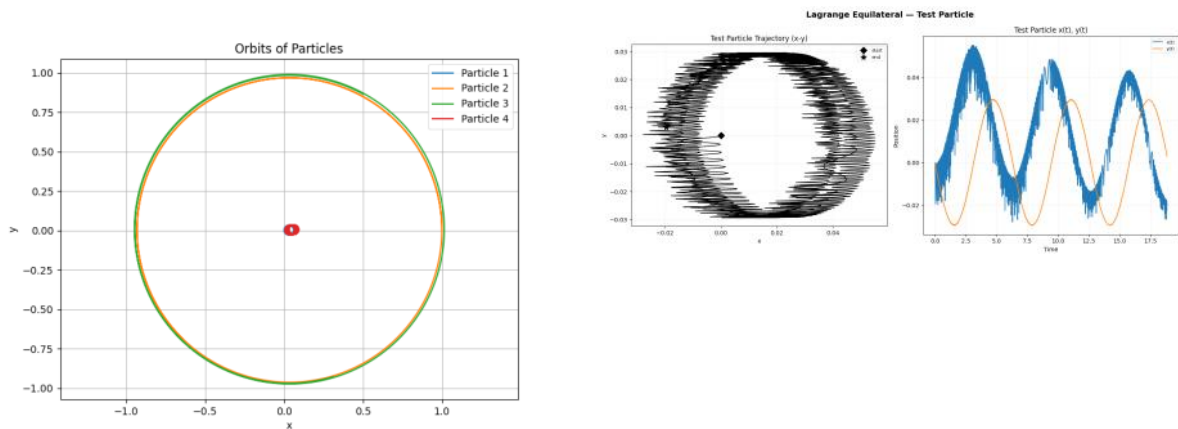
3c) Routing criteria

$$M = [1, 0.03, 0.003] / M$$

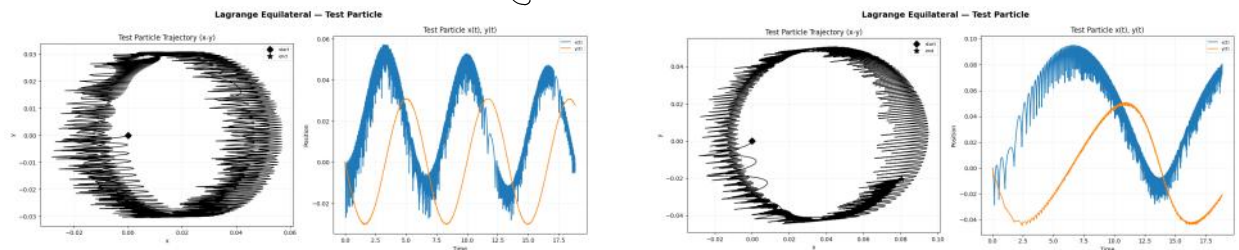
This plot shows the non-perturbed system. The fourth body stays close to the largest body.



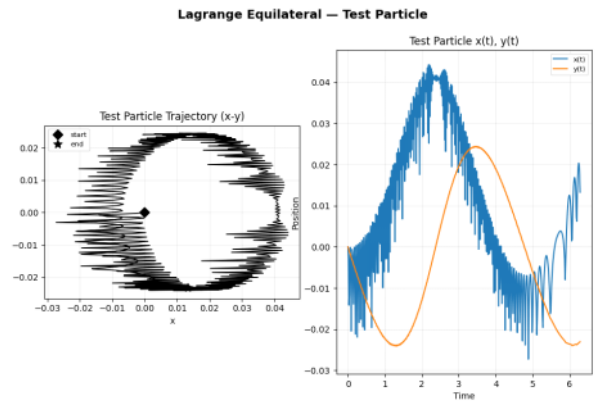
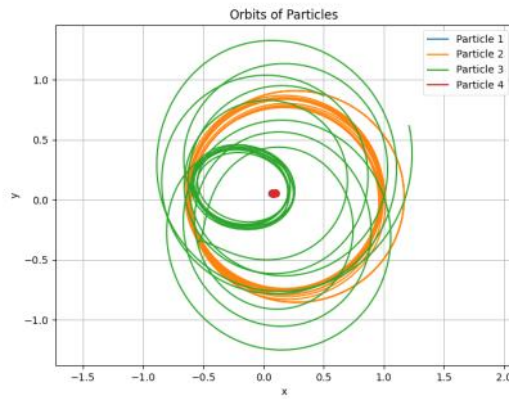
This plot shows the system by 0.001 in the x & y directions of body 1 (nearest). The perturbation is not enough to impact the stability of the orbit. The fourth body stays close to m_1 throughout the period.



These two plots show the fourth body trajectories with a perturbation of 0.01 (left) and 0.1 (right) in the x & y direction of m_1 . The body continues to stay close to m_1 .



These plots show the system perturbed by 0.5 in the x & y direction of m_1 . The third body's orbit becomes unstable but still does not escape. The fourth body continues to stay in orbit around m_1 with influences on its trajectory by the other bodies.



Adding a fourth body to these system that does not impact the underlying dynamics & plotting its trajectory shows the complex interactions & forces between the bodies. The attractive forces are complex & hard to predict. Introducing perturbations accelerates orbit instability & escape of the smallest body.

Systems that meet the Routhy criteria are stable until large perturbations are introduced. However, it takes many periods for the impact to the body trajectories to become apparent. It is possible to force a Routhy system to become unstable if the perturbation is "large".